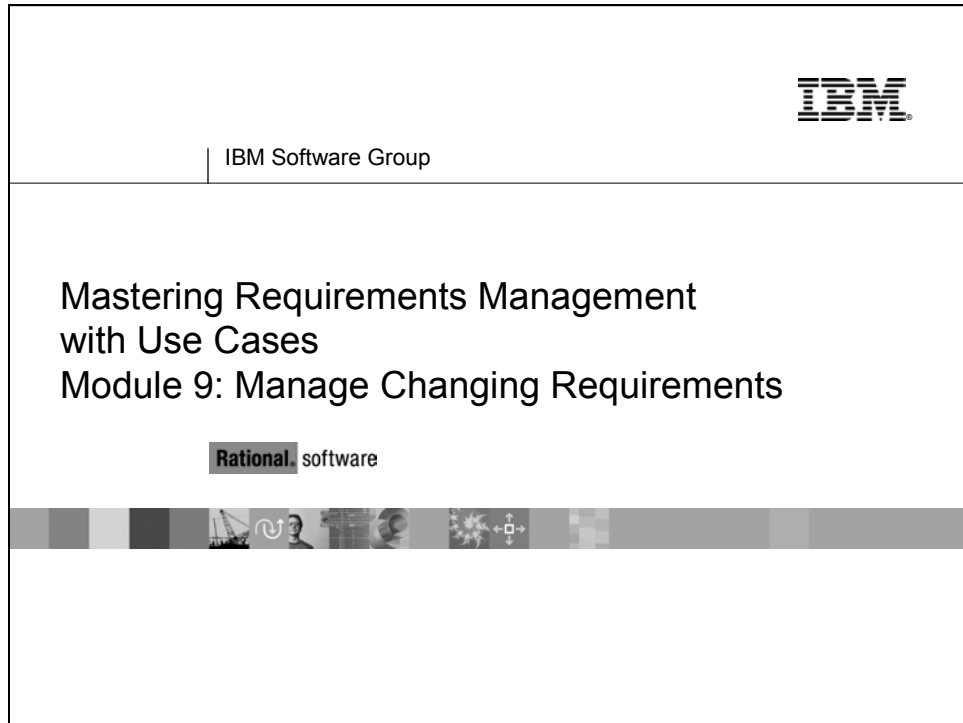


► ► ► **Module 9** **Manage Changing Requirements**



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Objectives

Objectives

- ♦ Describe techniques for managing change.
- ♦ Gain control of the requirements' input, output, and process.
- ♦ Use a requirements repository.
- ♦ Measure status and change periodically.
- ♦ Trace requirement relationships through the lifecycle.
 - Follow a single channel for approval of requests.
- ♦ Use configuration management.

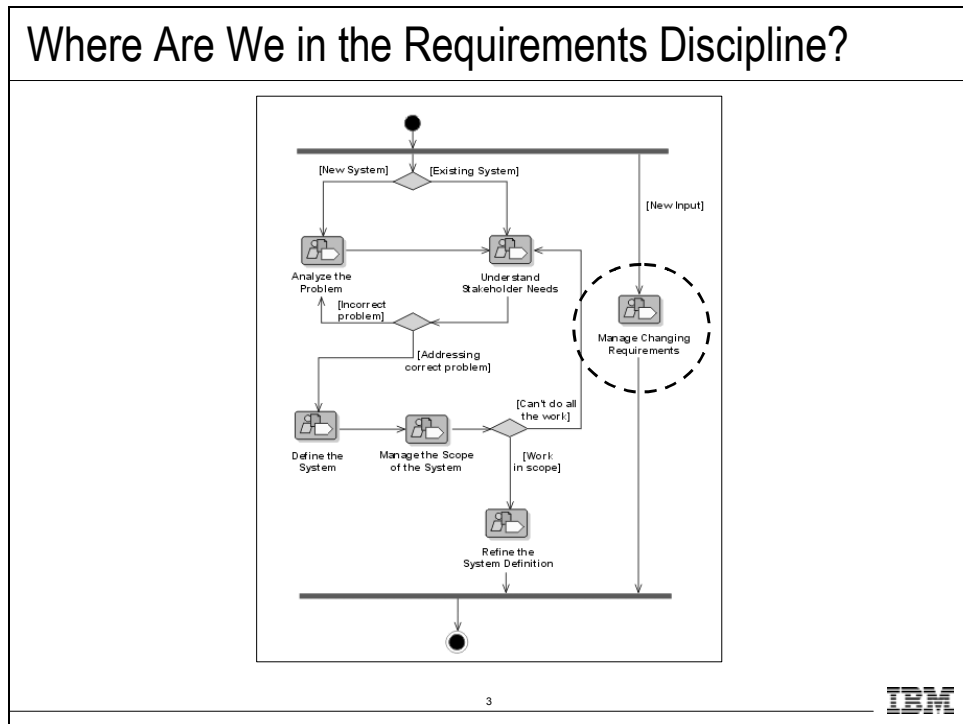
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In this module you discuss various ways to manage changing requirements.

An iterative process allows you to refine the model in smaller increments at each iteration and helps us maintain control of the changes throughout the product lifecycle.

Where Are We in the Requirements Discipline?

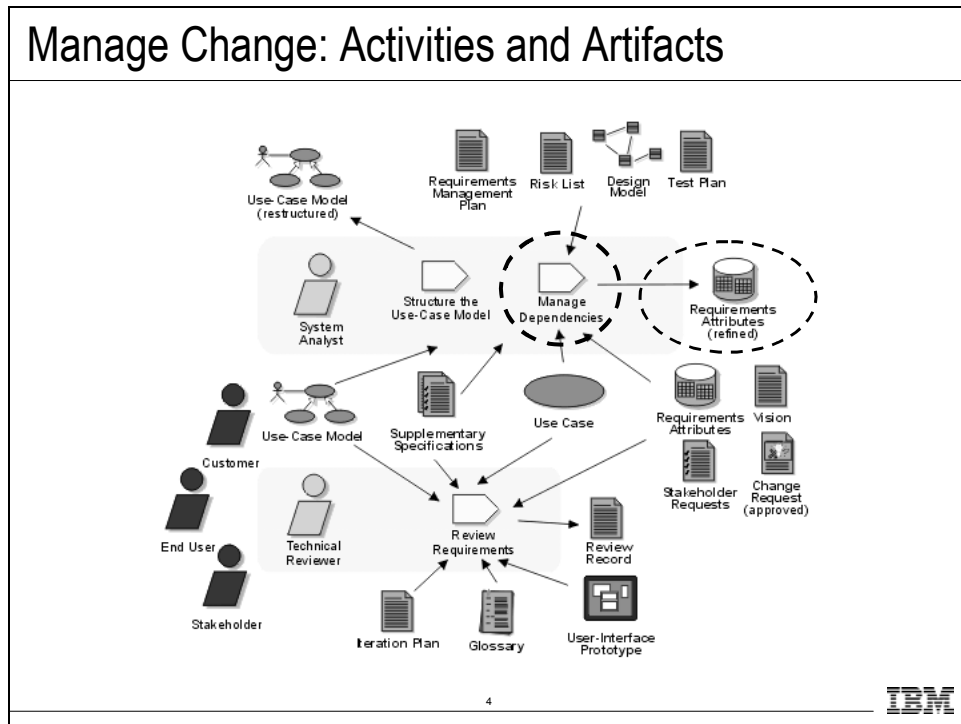


Where does Managing Changing Requirements workflow detail fit in the development process?

The workflow detail called “Manage Changing Requirements” in the Rational Unified Process® represents activities you might undertake to help you manage change throughout the lifecycle of the system’s development.

Where is this done in your own software development process?

Manage Change: Activities and Artifacts



The purpose of this workflow detail is to:

- Evaluate formally submitted change requests and determine their impact on the existing requirement set.
- Structure the use-case model. (Covered in Module 10.)
- Set up appropriate requirements attributes and traceability.
- Formally verify that the results of the Requirements discipline conform to the customer's view of the system.

Changes to requirements naturally influence the models produced in the Analysis & Design discipline, as well as the test model created as part of the Test discipline.

Traceability relationships between the requirements identified in the manage dependency activity of this discipline and others is the key to understanding these effects.

Another important concept is the tracking of requirement history. By capturing the nature and rationale of requirement changes, reviewers (in this case, the role is played by anyone on the software project team whose work is affected by the change) receive the information needed to respond to the change properly.

Why Do Requirements Change?

Why Do Requirements Change?

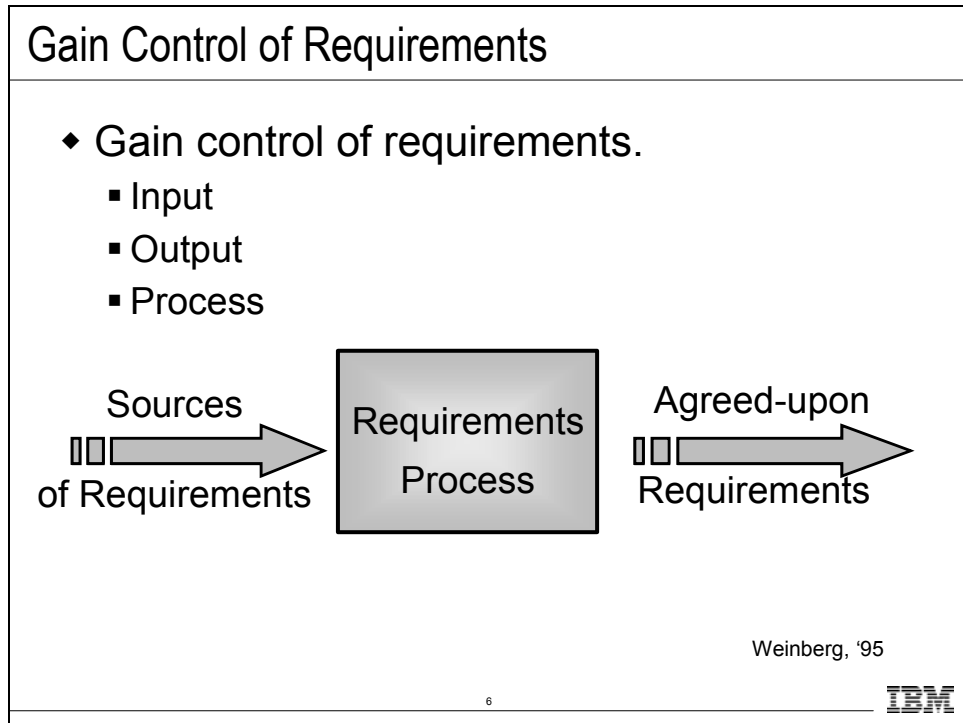
- ♦ Our understanding of the problem improved.
- ♦ The problem being solved changed.
- ♦ We failed to ask the right people the right questions at the right time.
- ♦ We failed to create or follow a process to help manage change.
- ♦ The users changed their minds or their perceptions.
- ♦ The external environment changed.

5



Have you ever worked on a project where there were **no** changes in the requirements from the initial agreement through the end of the project?

Gain Control of Requirements



Why is scope management so hard? It is because the requirements change and other environmental factors (such as resources, budget, and so on) change around us.

So, you need to have an ongoing process that assures the team that the amount of work you are planning to do stays doable and under control.

Gain Control of the Requirements Input

Gain Control of the Requirements Input	
Official Sources ▪ Requirements specifications ▪ Customer requests ▪ Marketing department ▪ Bug reports and enhancement requests ▪ Competitive analysis	Unofficial Sources (Leakage) ▪ Enhancements mentioned by distributors at a convention ▪ Direct customer requests to programming ▪ Inserted by programmers with “careful consideration” of what’s good for the customer ▪ Shipped mistakes that must be supported ▪ Hardware features that did not get in or do not work ▪ Knee-jerk change of scope reaction to competitors ▪ Programmer’s “Easter Eggs”
Weinberg, '95	7 

Note: The actual sources (people) are the same for both columns, such as customers, marketing, and developers.

What is the difference between the two columns? Adherence to an agreed-upon process, rather than ad hoc, uncontrolled input. The sources are not the problem, but they must be handled within a process.

What do you do about unofficial sources?

It is important to understand that all of your stakeholders have a vested interest in the outcome of the system. As such they are valid sources of requirements. The important thing is to recognize that and treat the feedback they provide in a controlled manner.

- Recognize that requirements come from many sources.
- Know your sources.
- Accept that the interests are legitimate.
- Create an explicit negotiation process.
- Develop a process that guarantees all requirements funnel through one channel.
- “Batch” requests.
- Consult CCB (Configuration Control Board).
- Involve product managers.

An Easter Egg is an unwanted feature that is hidden deep inside an application. It is usually activated by an obscure sequence of keystrokes. For example, the developers at Microsoft buried a flight simulator inside Excel 97.

Gain Control of the Requirements Output

Gain Control of the Requirements Output

- ◆ Document the requirements.
 - Write them down.
 - Collect requirements in a central repository.
 - Make requirements accessible to all stakeholders.
 - Keep them reasonably stable.
 - Control the changes (avoid “scope creep”).
- ◆ Use requirement attributes and traceability.
 - Gather useful information from the right people.
 - Attributes/traces in a repository along with requirements.

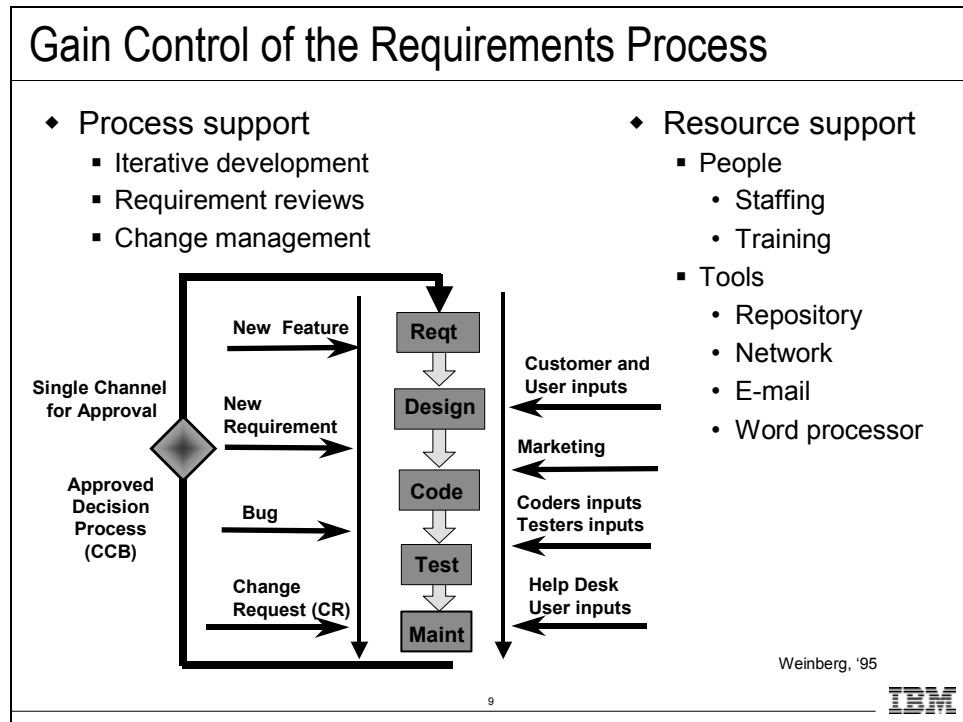
Weinberg, '95

8



The key is to keep the requirements visible. Write down what is agreed upon and accessible to all team members.

Gain Control of the Requirements Process



It is important to provide support for requirements management, including both process and resource support.

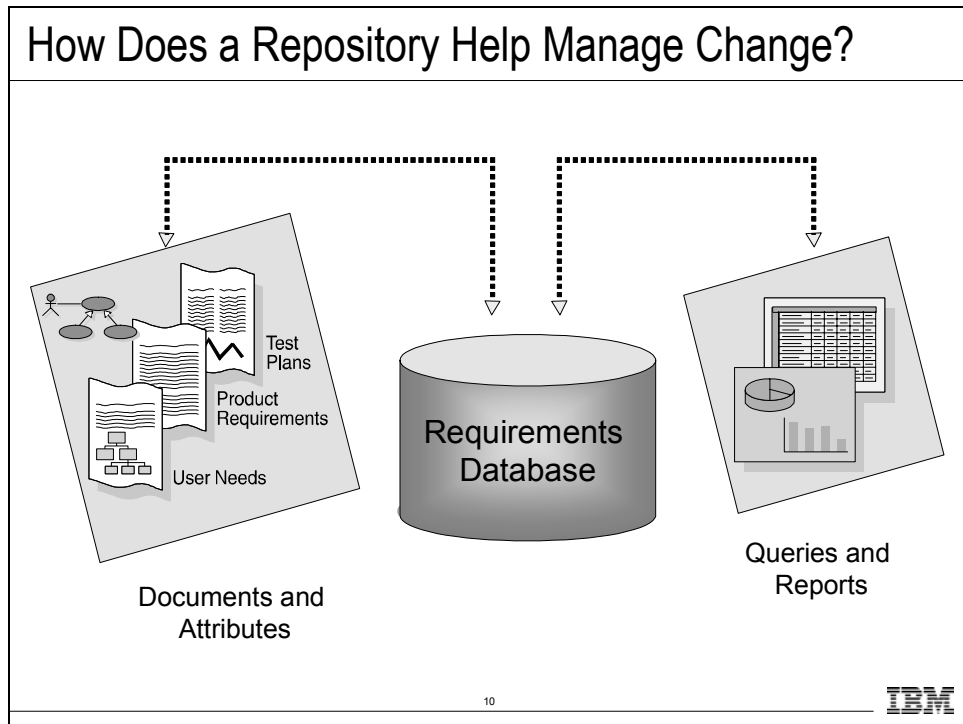
As defects and enhancement requests (hereafter both called a *change request*) come in during our lifecycle, intercept them and pass through a single approval channel.

The single channel of approval may be one person (such as a project manager) or a group of representatives (Change or Configuration Control Board (CCB)). The CCB should have representatives from each of the relevant groups.

The CCB examines each change request and determines the appropriate course of action. The course of action may be to simply fix a bug because a requirement was not implemented correctly, or it may involve changing an existing or creating a new requirement.

If the change request involves a change to the requirements set, the change request is used as input to writing the requirement. It does not become the requirement. Remember, requirements must be unambiguous, consistent and verifiable. Change requests are typically not written with these measures in mind.

How Does a Repository Help Manage Change?



Here is architecture that may be used for organizing documents and maintaining project requirements and attributes in a central database. The database is used for collecting metrics, generating reports, and answering queries relating to requirements within the project.

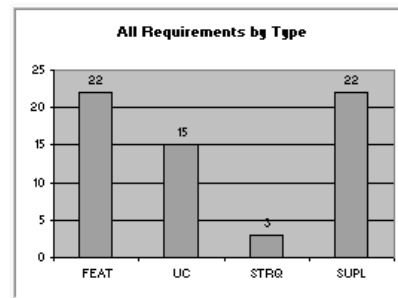
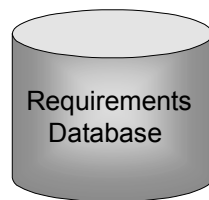
Traceability, which is a very powerful tool for managing change, is very difficult to do manually. A database helps manage change by automating traceability: links between requirements are automatically maintained, and it is easy to find other requirements linked to a particular requirement.

A central database also helps manage change by automating the collection and reporting of statistics about requirements. For example, a repository should be able to answer questions, such as the number of requirements that have not yet been approved.

Metrics Help Manage Change

Metrics Help Manage Change

- ♦ Create statistical reports for project management.



Project Metrics
(MS Excel)

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A repository provides the ability to compile statistics, or metrics, about the project requirements, requirement attributes, relationships, and revisions. These metrics can be used to help manage changing requirements.

Project managers and system analysts can use metrics to assess project status, progress, and expected performance. Meaningful data can be dispersed to team members, management, and customers.

Use requirement metrics to retrieve information for evaluating the status and changes in a project: priorities, workloads, the addition of new requirements, and changing or unstable requirements.

What Metrics Help Manage Requirements?

What Metrics Help Manage Requirements?

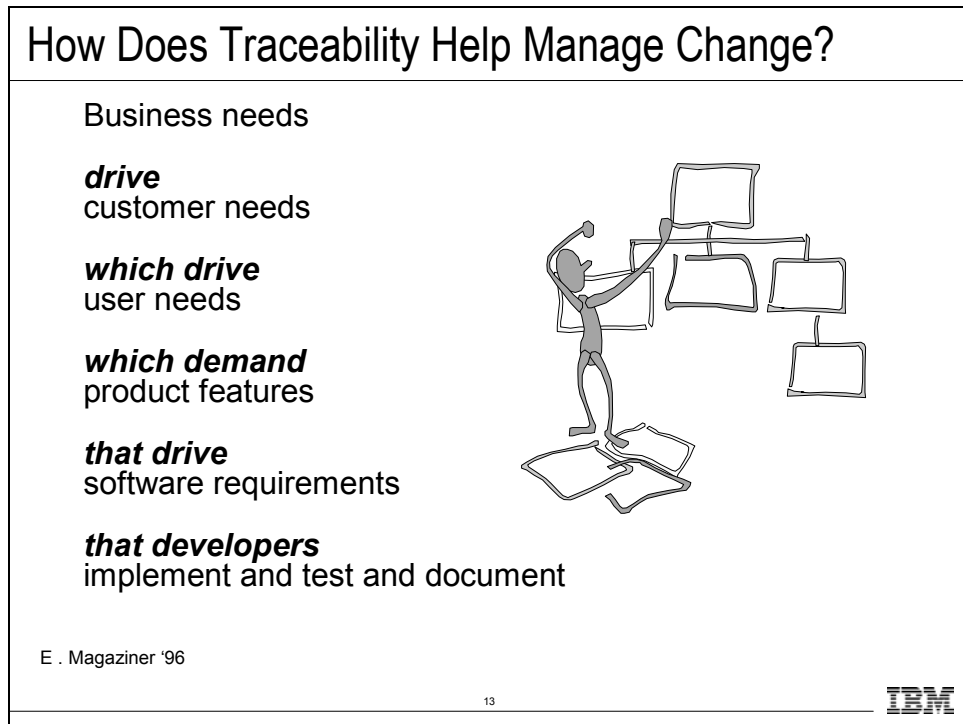
- ♦ How many use cases do we have?
- ♦ What critical features haven't been approved?
- ♦ What is the estimated cost of the proposed changes?
- ♦ How many changes since the last review?
 - Who authorized the change?
 - What is the impact of the change on the testing schedule?
 - Are there fewer changes this week than last week?

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The white paper, *Metrics for Requirements Management*, has more discussion about requirements metrics for project management.

How Does Traceability Help Manage Change?



Tracing requirement links one requirement to other related requirements.

Traceability helps ensure that you are building the right system. Tracing from one requirement to related requirements also helps ensure that if a requirement is changed, then related requirements also reflect that change.

For example, if a customer no longer needs a particular part of the problem solved, then what features, software requirements, and test requirements can be eliminated?

The slide presents some of the different levels of requirements you may wish to trace.

Why Use Requirements Traceability?

Why Use Requirements Traceability?

- ◆ **Assure quality**
 - Verify that the implementation fulfills all requirements.
 - Verify that the application does only what was intended.
- ◆ **Analyze impact of a change**
 - Find related requirements.
 - Inspect related requirements.

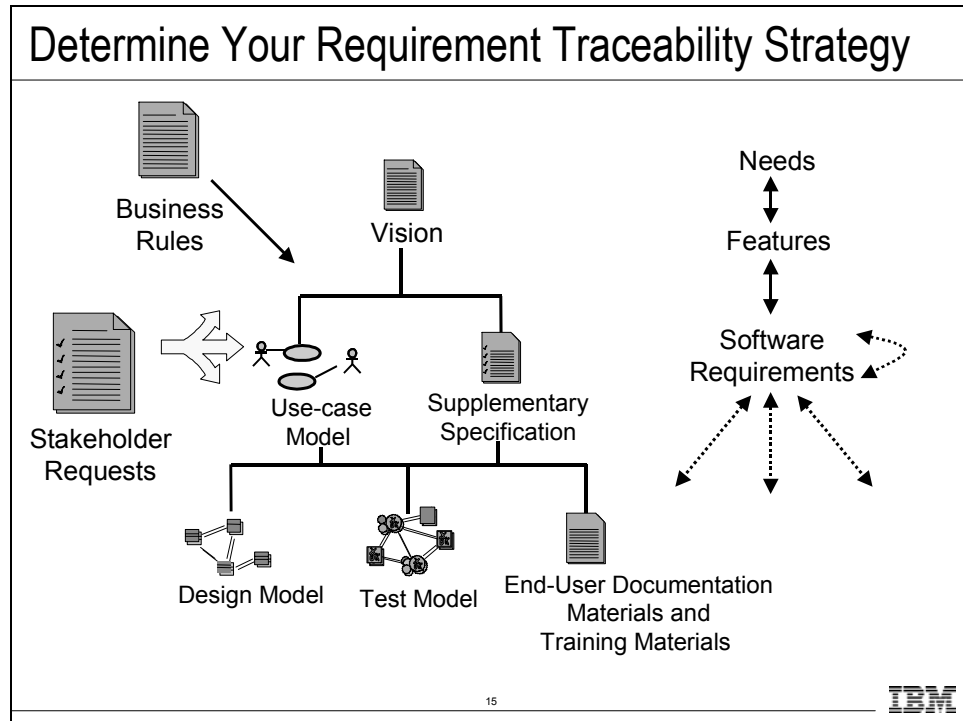
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One of the keys to effective change management of requirements is traceability of requirements to other project elements.

Requirements traceability is a proven technique for assuring quality. The technique is recommended by standards and assessment groups, such as IEEE and CMM. It is also mandated by many organizations, such as the Department of Defense (DoD), and the Food and Drug Administration (FDA).

Determine Your Requirement Traceability Strategy



The first step is to establish your requirements structure and the relationship of different requirements types to each other.

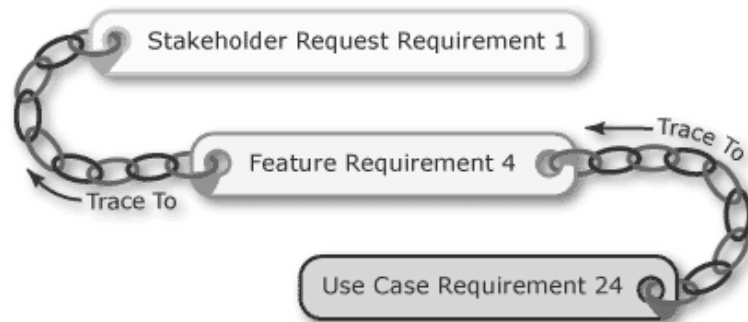
Based on this structure, you need to set up traceability links between all associated requirements or other project elements.

- Trace top-level requirements to detailed requirements.
- Trace requirements to design.
- Trace requirements to test procedures.
- Trace requirements to user documentation plans.

Trace Requirements to Each Other

Trace Requirements to Each Other

- ♦ Traceability links verify that all lower-level derived requirements are consistent with stakeholder needs.



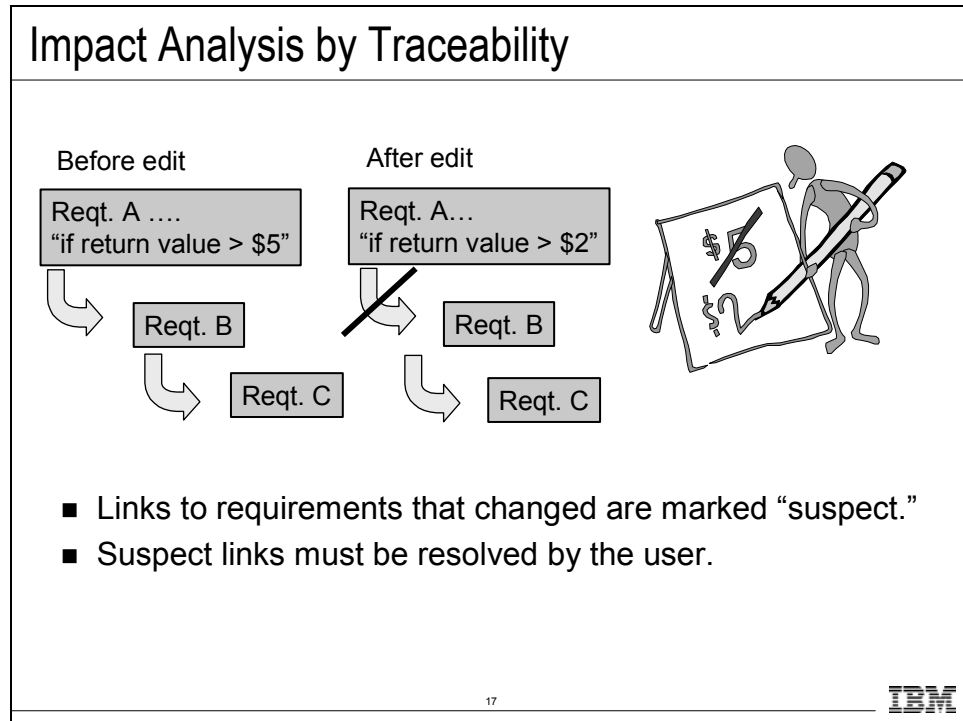
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Here are some types of links that may be appropriate in our requirements.

You may trace a feature directly to a use case. By tracing a feature to a use case, you can verify that each use case is derived from a feature that is directly related to a stakeholder need.

Impact Analysis by Traceability



RequisitePro provides what are called "suspect links," which can notify that an associated requirement has changed.

All directly related requirements should be reviewed to assess whether they are affected.

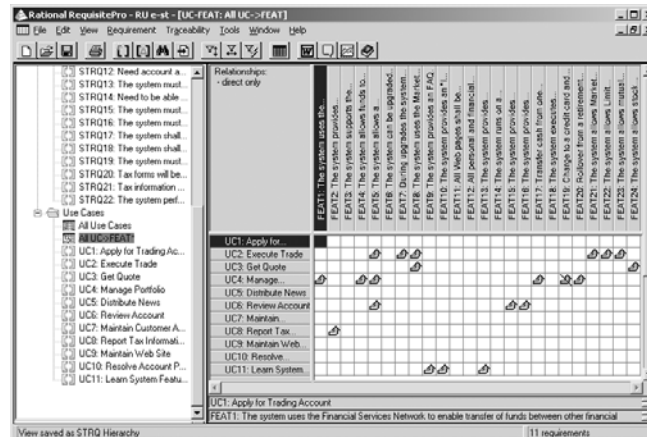
Why is the link from Requirement **B** to Requirement **C** not marked as suspect?

These suspect links can only be resolved manually by actually looking at the changes and the affected requirements.

View Relationships: Traceability Matrix

View Relationships: Traceability Matrix

- ♦ Shows relationship between two requirements.



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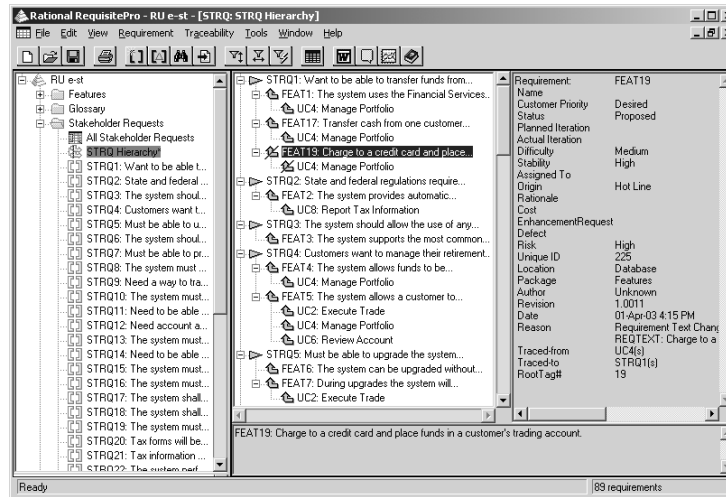
Here is a way you might view the traceability links using a requirements management tool, such as Rational RequisitePro®.

A traceability matrix presents a two-dimensional view of the traces from one requirement type to the same requirement type or another requirement type.

View Relationships: Tree Report

View Relationships: Tree Report

- ◆ Shows the full hierarchy of traceability links.

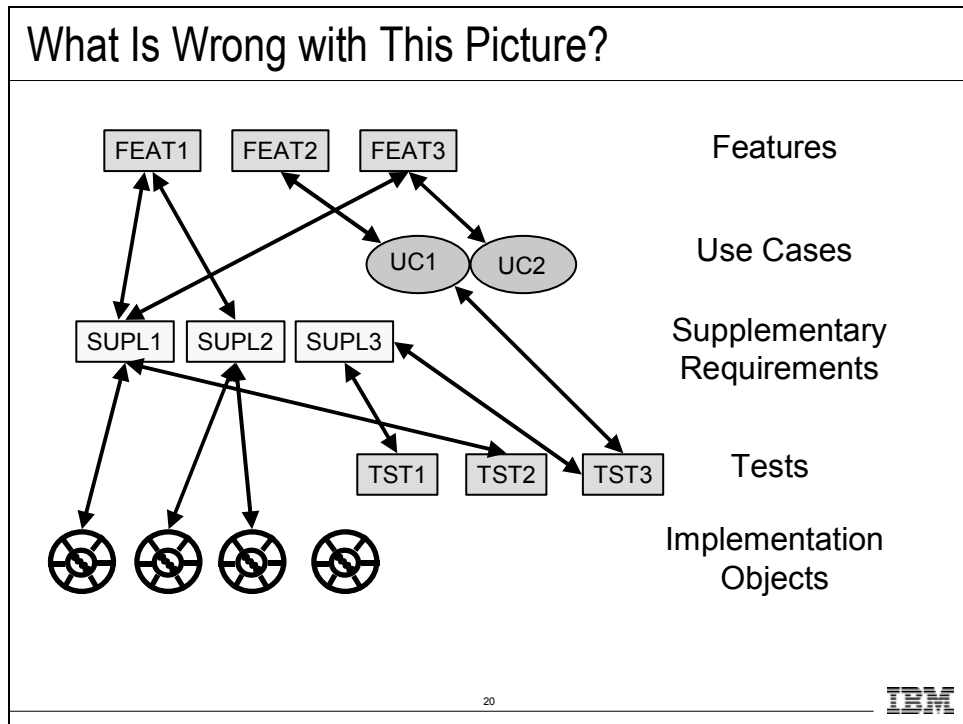


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The Tree view expands a particular requirement or set of requirements so that you can see its full depth of traceability.

What Is Wrong with This Picture?



All top-level requirements should trace to a software requirement (either within a use case, a supplementary spec, or a formal SRS) and eventually to tests and implementation objects.

Traceability Queries Help Manage Change

Traceability Queries Help Manage Change

- ◆ Requests not linked to product features.
- ◆ Status of tests on use cases in Iteration #3.
- ◆ Supplementary requirements linked to tests with status of “untested.”
- ◆ Results of failed tests in order of criticality.
- ◆ Features scheduled for this release, the stakeholder requests they satisfy, and their status.

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Traceability can supply managers and project leaders with valuable information for managing changing requirements.

What types of questions might you ask that could be answered by knowing about the links between the requirements?

Discussion: Traceability

Discussion: Traceability

- ♦ How would you link each feature to a use case or supplementary specification in our requirements?
- ♦ What should we do about requirements that don't trace?

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Look at each of the stakeholder requests and features (from our exercises 5.1 and 6.1) for our class project and set up a sample traceability for a few of them through your use-case model.

Configuration Management in Requirements

Configuration Management in Requirements

- ◆ Prevent unauthorized change.
- ◆ Preserve revisions of requirements documents.
- ◆ Retrieve previous versions of documents.
- ◆ Allow a managed baseline release strategy.
- ◆ Prevent simultaneous update of documents.

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It is important to keep track of the change history of requirements and the requirement documents. Configuration management of requirements is important for the same reasons that configuration management of source code is important.

Manage Changing Baselines

Manage Changing Baselines

When do you **freeze** the specification?

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Do you “freeze” your requirements specifications? If so, when? What does this mean?

Review: Managing Changing Requirements

Review: Managing Changing Requirements

1. What are some reasons requirements change?
2. How does a Change Control Board (CCB) help with scope creep?
3. What are benefits of a requirements repository?
4. How can requirements traceability help manage change?
5. What are some benefits of keeping your requirements under configuration management?

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